

Mounting

Filesystems are usually used by being **mounted** - associated with a directory
This can be done **temporarily** or **persistently** across reboots

Temporary Mounting

mount - mount filesystem

umount - unmount

mount

```
mount [-o options] [-t type] [device] [mountpoint]
```

type: filesystem type e.g. *ext4*

device: device you want to mount e.g. */dev/sdb1*

mountpoint: directory to attach the device e.g. */mnt/something* is common options

-r	mount as read-only
-w	mount as read and write (default)
-L	mount with a custom label

example

```
mount /dev/sdb1 /mnt/shared
```

autodetects the filesystem type

umount

```
umount [options] [-t type] [device | mountpoint]
```

Ensures any cached info is written to the device before it's removed

Using a GUI's unmount or eject options is equivalent to using *umount*

Permanent Mounting

/etc/fstab contains static file system information

1. Identify the device UUID (Universally Unique Identifier)

```
sudo blkid
```

2. Create mount point

```
mkdir /mnt/mydata
```

3. Add device to **/etc/fstab**

```
# <file system> <mount point> <type> <options> <dump> <pass>
UUID=your-UUID-here /mnt/mydata ext4 defaults 0 2
```

options: defaults is a shorthand that includes rw, exec, auto etc

dump: used by dump backup utility (0 means disabled)

pass: Determines the order of filesystems to check at boot time

0 - no check

1 - root filesystem

2 - other filesystem

4. **sudo reboot**

backup /etc/fstab before editing in case of error

after editing /etc/fstab check for errors `sudo mount -a`